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Genetic algorithm approach for precedence-constrained sequencing problems

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Abstract In this paper we propose a genetic algorithm (GA) approach based on a topological sort (TS)-based representation procedure for effectively solving precedence-constrained sequencing problems (PCSPs). The TS-based representation procedure used in the proposed GA approach can generate feasible sequences in PCSPs. By applying the proposed GA approach, the sequence determination problems with precedence constraints can be easily solved. Experimental results show that the proposed GA approach is a good alternative in locating optimal sequence for various types of PCSPs.

Keywords Genetic algorithm · Precedence-constrained sequencing problems · Mixed integer programming · Topological sort

Introduction

Precedence-constrained sequencing problem (PCSP) can be applied to a number of economic problems such as project management, logistics, routing, assembly flow, scheduling, and networking. Finding out an optimal sequence for PCSP may cause a great economic effect because infeasible sequence determination highly affects all downstream stages of manufacturing, logistics, and networking.

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PCSP is to locate the optimal sequence with the shortest traveling time among all feasible sequences, visiting each node only once on a directed graph $G(C, A)$, where $C = \{1, 2, \ldots, I\}$ is a set of nodes, and $A = \{\langle i, j\rangle, i, j \in C$ and $i \neq j\}$ is a set of precedence relations. Precedence relation $\langle i, j\rangle$ determines the sequence of traveling between given pair of nodes. This problem can be formulated as a traveling salesman problem (TSP) with precedence constraints.

A number of industrial models for PCSP have been constructed. Chen (1990) used an *AND/OR* PCSP to solve assembly scheduling problems. In this model, the precedence relations result from geometric and physical constraints of the assembled items. He and Kusiak (1992) presented a PCSP and a heuristic algorithm to solve the scheduling model with sequence-dependent changeover cost and precedence constraints. Savelsbergh and Sol (1995) presented a PCSP model to solve the Dial-A-Ride problem where a vehicle should transport a number of passengers, and each passenger should be transported from a given location to a specific destination point. Renaud et al. (2000) suggested a heuristic model to solve a pickup and delivery TSP, which can be formulated as a PCSP model.

More recently, Moon et al. (2002) presented a genetic algorithm (GA) with a priority-based encoding method to solve the PCSP based on supply chain planning model. Duman and Or (2004) developed an approach to solve a PCSP in a printed circuit board assembly problem. This approach initially ignores precedence relations and solves the problem as a pure TSP, and then it is applied to eliminate component damage in the resulting TSP tour. Lambert (2006) formulated a disassembly scheduling problem by modification of the two-commodity network flow model, which is included in a PCSP. Su (2007) proposed a unique reasoning method supported by the artificial intelligent technique of case-based reasoning with evolutionary algorithm to solve a PCSP.

The conventional approaches mentioned above show that PCSP is a useful tool for a variety of industrial planning and scheduling problems (Chen 1990; Duman and Or 2004; He and Kusiak 1992; Moon et al. 2002; Moon and Seo 2005). However, it can be also seen that most approaches may not solve various types of PCSPs efficiently because of computational complexities and complicated precedence constraints. Also, computation time required greatly increases with the increase of the number of nodes. Accordingly, a new approach is needed. This new approach should easily handle the complicated precedence constraints and find an optimal sequence efficiently regardless of the increase of the number of nodes.

This paper aims at formulating a mathematical model as well as at developing an efficient GA approach with topological sort (TS)-based representation procedure for solving various types of PCSP. The TS-based representation procedure is applied to generate a set of feasible individuals for the population of GA approach within a reasonable computation time. The proposed mathematical model and the proposed GA approach have the following features.

- The proposed mathematical model is formulated by modifying the standard integer programming for TSP to avoid computational difficulties and ambiguity.
- Since the precedence constraints between nodes are handled by the TS-based representation procedure, the TS-based representation procedure produces only feasible sequences. Therefore, for the proposed GA approach, the constraint handling strategies, such as rejecting, repairing, modifying, or penalizing are not needed.
- Since the TS-based representation procedure produces only feasible sequences, the proposed GA approach is very efficient in terms of search time and space.

Mathematical formulation

PCSP requires that there is no cycle in precedence relations. It is clear that a linear sequence is impossible if the graph has a loop, because a sequence from i to j and from j to i is possible for two nodes i and j on the cycle. A feasible sequence passes through each node in the given directed graph only once and cannot visit any two nodes at the same time. Therefore, PCSP is a type of directed acyclic graph. Since a directed graph may has several feasible sequences, it is necessary to decide an optimal sequence among all feasible ones. The optimal sequence, which would minimize the total traveling time, can be decided by comparing all feasible sequences. In order to avoid computational difficulty and ambiguity, the proposed mathematical model is formulated by modifying the standard integer programming for the TSP (Baker 1974; He and Kusiak 1992; Pinedo and Chao 1999). The following notation is used to describe PCSP.

- i, j node index, $i, j = 1,\ldots, I$, where I is the number of nodes.
- t_{ij} traveling time from node i to j.
- A_i arrival time at node i.
- P set of nodes (i, j), where node i is visited before node j.
- R set of nodes (i, j), where nodes i and j can be visited in any feasible sequence.
- L an arbitrary large positive number.

A variable is introduced to adapt PCSP as follows:

$$y_{ij} = \begin{cases} 1, & \text{if node } i \text{ is visited before } j, \\ 0, & \text{otherwise.} \end{cases}$$

Let E be the total traveling time for visiting all nodes defined by Eq. 1.

$$E = \max_{\forall i} \{A_i\} \tag{1}$$

A mixed-integer programming model of PCSP for minimizing the total traveling time is as follow:

Minimize E

subject to

$$A_j - A_i \geq t_{ij} \quad \forall (i, j) \in P \quad \text{and} \quad i \neq j \tag{2}$$

$$A_i - A_j + L\, y_{ij} \geq t_{ij} y_{ij} + t_{ji} y_{ji}, \quad \forall (i, j) \in R \quad \text{and} \quad i \neq j \tag{3}$$

$$y_{ij} + y_{ji} = 1, \quad \forall (i, j) \in R \quad \text{and} \quad i \neq j, \tag{4}$$

$$y_{ij} = 0, 1 \quad \forall (i, j) \in R \quad \text{and} \quad i \neq j \tag{5}$$

Constraints (2) and (3) ensure that two nodes cannot be visited at the same time. Constraint (4) represents that any two nodes are to be visited in only one sequence. This constraint represents the technical restriction on visiting sequence as the existence of a partial sequencing among the nodes based on precedence relations. Also, this constraint represents that there is no cycles in precedence relations. Constraint (5) means integer variable.

GA approach

For implementing the proposed GA approach, the detailed procedure is appeared in the following subsections. First, TS-based representation procedure considering precedence constraints is suggested. By the use of the TS-based representation procedure, the initialization process of GA population is followed. Two representation procedures, priority-based representation procedure and TS-based representation procedure, are suggested and compared, respectively. Secondly,

Table 1 Precedence matrix

	Node number				
	1	2	3	4	5
Node number					
1			1	1	
2				1	
3					1
4					1
5					

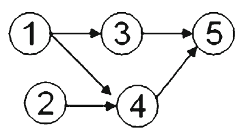

Fig. 1 A directed graph with 5 nodes with precedence constraints

a fitness test for the individuals resulting from GA search process is done. Selection, crossover and mutation operators are used for genetic operator.

TS procedure with precedence constraints

Generally, a precedence relation $\langle i, j\rangle$ is called a partial sequencing between two nodes i and j. Also, TS procedure requires that there are no cycles, which would determine that any node i should be visited before itself in any sequence. To represent precedence relations, we can use the precedence matrix $D = [d_{ij}]$, where

$$d_{ij} = \begin{cases} 1, & \text{if node } i \text{ is visited immediately before } j, \\ \text{null}, & \text{otherwise.} \end{cases} \quad (6)$$

where $d_{ij} = 1$ indicates the precedence relation $\langle i, j\rangle$.

The precedence matrix of the example with 5 nodes and the precedence relations is shown in Table 1.

For example, $d_{13} = 1$ in Table 1 means that node 1 should be visited immediately before node 3, that is, node 1 precedes node 3 in linear sequencing. We call node 1 an immediate predecessor of node 3, and node 3 is an immediate successor of node 1. On the other hand, if d_{ij} = null, then there is no relationship between two nodes i and j. However, if $d_{23} = 1, d_{34} = 1$, and $d_{24} = 1$, then the graph cannot generates a feasible sequence because it has a cycle. Also, in the matrix columns 1 and 2 of Table 1 have 'null' values. It indicates that nodes 1 and 2 have no predecessors. From the precedence constraints in Table 1 we can draw a directed graph shown in Fig. 1.

The TS procedure used for producing a feasible sequence on a directed graph is described in Fig. 2.

The TS procedure in Fig. 2 is a key role of generating a set of feasible sequences on the directed graph.

```
procedure: TS procedure
input: a precedence matrix and its directed graph
    I: number of nodes to visit
output: feasible sequence on the directed graph
step 0: for all nodes
            If all nodes have a predecessor or there is no
              node i;
            then the directed graph is infeasible, stop;
            else randomly select a node i without any
                predecessors;
                 return the node i;
                 remove the node i and its arc from the
                 direct graph;
          end if
      end for
```

Fig. 2 TS procedure

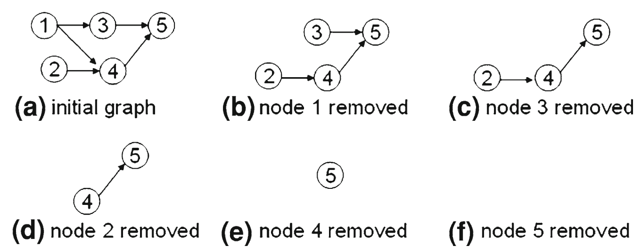

Fig. 3 Priority-based representation procedure

Representation and initialization by TS procedure

In order to generate a feasible sequence, the representation scheme should consider the precedence constraints of the given PCSP. Therefore, developing the representation scheme is an important issue in designing GA approach. The most common representation scheme for directed acyclic graph is the priority-based representation procedure (Gen and Cheng 2000; Gen et al. 2006; Moon et al. 2002; Moon and Seo 2005). This scheme has been widely used to represent feasible sequences. It has, however, serious limitations such as idling. For example, if we consider the directed graph of Fig. 1, then the corresponding node priorities can be randomly generated as follow:

$$(4, 1, 5, 2, 3) \quad (7)$$

where the values of each gene are within $(1, I)$ exclusively.

The value of each gene indicates the priority for candidate selection and the selection sequence is based on the nodes without incoming arcs. The feasible sequence produced by the priority-based representation procedure is shown in Fig. 3.

In Fig. 3, nodes 1 and 2 have no predecessors and their corresponding priority values are 4 and 1, respectively. Node 1 is selected first for a feasible sequence because its priority is higher than that of node 2. After selecting node 1, we can remove it along with all outgoing arcs as shown in the situation (b) of Fig. 3. Now nodes 2 and 3 have no predecessors and node 3 is selected because its priority is higher than that

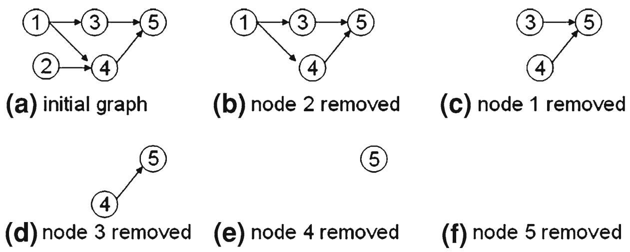

Fig. 4 TS-based representation procedure

of node 2. Accordingly, the feasible sequence for all nodes can be obtained as follows:

$$\langle 1 \quad 3 \quad 2 \quad 4 \quad 5 \rangle \tag{8}$$

However, there are some difficulties in producing various types of feasible sequences. For example, in the nodes 1 and 2 of the situation (a) of Fig. 3, another feasible sequence $\langle 2 \quad 1 \rangle$ can be produced, If the priority constraint does not be considered. Similar case is also happened in nodes 2 and 3 of the situation (b) of Fig. 3. Finally, the various types of feasible sequence do not be produced because of the priority constraint assigned in each gene. This case is called idling and restricts the efficiency of the search process. Because of idling, the priority-based representation procedure cannot guarantee optimal solution within reasonable generations or computation time.

In order to overcome the difficulties, an efficient and simple representation scheme using TS procedure, called TS-based representation procedure, is proposed. This procedure does not use any priority in each node, when comparing with priority-based representation procedure. For example, if we consider the directed-graph shown in Fig. 1, a feasible sequence by using the TS-based representation procedure can be produced. The detailed procedure is shown in Fig. 4.

In Fig. 4, a node between the nodes 1 and 2 without any predecessors is randomly selected. If the selected node is 2, it is removed in the directed graph and the remainders are represented as the situation (b). Secondly, only node 1 should be selected and removed as shown in the situation (c). The situations (d), (e) and (f) can be also represented as the same way of the situations (b) and (c). Finally, a feasible sequence is produce as follow:

$$\langle 2 \quad 1 \quad 3 \quad 4 \quad 5 \rangle \tag{9}$$

This feasible sequence can be used in the individual for GA population

The next step is to produce the initial population for implementing GA approach. By the TS-based representation procedure, we can easily produce some feasible sequence sets for the individuals as many as the population size, *pop_size*.

If *pop_size* is 3, the initial population can be generated as shown in Fig. 5.

Fig. 5 Initial population

$$v_1 = [2 \mid 1 \mid 3 \mid 4 \mid 5]$$
$$v_2 = [2 \mid 1 \mid 4 \mid 3 \mid 5]$$
$$v_3 = [1 \mid 3 \mid 2 \mid 4 \mid 5]$$

Fitness test and selection

To improve the performance of the individuals resulting from GA search process, each individual should be evaluated by measuring its fitness. The fitness value of each individual is computed and the objective is to find an individual with the shortest traveling time. We consider the objective function of mixed-integer model which is to minimize total traveling time. Let $f(k)$ be the fitness function of k-th individual, then $f(k)$ can be expressed as follows:

$$f(k) = \min\{\max_{\forall i}\{A_i\}\} \tag{10}$$

Selection strategy is to choose the respective individuals from current population, and the chosen respective individuals are considered as the population of next generation. For selection, the elitist selection strategy in enlarged sampling space (Gen and Cheng 1997) is used.

Crossover operator

Crossover operator is about exchanging genes among individuals to improve solution quality through GA search process. It works by combining subsequences from different parent individuals to generate a new one. However, some general types of crossover operator are not used for PCSPs, since most of the individuals resulting from crossover operators will violate the precedence constraints. Therefore, a new procedure of crossover operator considering PCSPs is proposed as follows:

Step 0: Randomly select an individual a from current population.
Step 1: Randomly select two positions b and c in the selected individual.
Step 2: Generate a random integer number d from (0, 2). If $d = 0$ then select the left subsequence of the first cut point, if $d = 1$ then select the sequence between first and second cut points, if $d = 2$ then select the right subsequence of the second cut point for crossover.
Step 3: Use TS-based representation procedure to produce an alternative subsequence for the segment selected in Step 2.
Step 4: Replace the subsequence selected in Step 2 with the alternative one in Step 3.

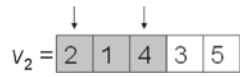

Fig. 6 Before crossover operator

Fig. 7 After crossover operator

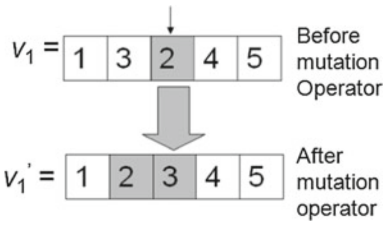

Fig. 8 An example of insertion operator

Steps 0 to 2 of the above procedure can be represented as the following string.

$$(a, b, c, d) \tag{11}$$

where the range of a is (1, *pop_size*), b and c are (1, I), respectively, and d is (0, 2).

Let's consider the 5 nodes with precedence constraints shown in Fig. 1 to implement the crossover operator. If $(a, b, c, d) = (2, 1, 3, 1)$, the selected individual and its segment is shown in Fig. 6.

For producing an alternative subsequence for the segment selected in Step 2, TS-based representation procedure is used in Step 3. Finally, the feasible subsequence $\langle 1 \quad 2 \quad 4 \rangle$ is produced. The new individual and its sequence for Step 4 are shown in Fig. 7.

Mutation operator

Mutation operator has a background role. Insertion operator (Gen and Cheng 1997) with TS-based representation procedure is used. The scheme randomly selects a gene within a selected individual and then inserts the gene in the position of its immediate predecessor or immediate successor. The detailed procedure is as follows:

Step 0: Randomly select an individual from current population.
Step 1: Randomly select a gene within the selected individual.
Step 2: Insert the gene in the position of its immediate predecessor or immediate successor with a feasible sequence using TS-based representation procedure.

An example of insertion operator is shown in Fig. 8.

Table 2 Performance measures

Measure	Description
CPU time	Average CPU time (in second)
Optimal value	Sum of traveling time for the feasible sequence with minimal traveling time among all feasible sequences
Optimal sequence	Feasible sequence with the minimal traveling time among all feasible sequences

Experiments

In this section, two types of PCSPs are considered. The first type uses a direct graph and a data set used in conventional approach to compare the performance of the proposed GA approach with that of the conventional approach. In the second types, three cases of PCSPs are considered and are used for comparing the performances between the TS-based representation procedure used in the proposed GA approach and the priority-based representation procedure used in conventional study (Moon et al. 2002).

For experimental comparison of the second types under a same condition, TS-based representation procedure and the priority-based representation procedure are used in the proposed GA approach, respectively. When the latter is used in the proposed GA approach, each step of the sections "Crossover operator" and "Mutation operator" for crossover and mutation operators is performed by the use of priority-based representation procedure instead of TS-based representation procedure. They are programmed in Visual Basic, and are run on IBM-comfortable PC Pentium IV 1.2GHZ processor, 1GB RAM and Window-XP.

The parameters used in the two approaches are set as follows: total generation number is 2,000, population size is 20, crossover rate is 0.5, and mutation rate is 0.05. Altogether 20 iterations are made to eliminate the randomness of the approaches.

Table 2 shows the measures of performance for comparing each approach.

In Table 2, the CPU time is averaged over 20 runs. The optimal sequence and the optimal value mean the best result when each algorithm reaches to a pre-defined total number of iterations (in our cases 20 runs).

Test problem 1 (T-1)

For T-1, the directed graph with precedence constraints and the corresponding traveling times used in the study of He and Kusiak (1992) are considered in Fig. 9 and Table 3, respectively. Using the data from Fig. 9 and Table 3, the proposed

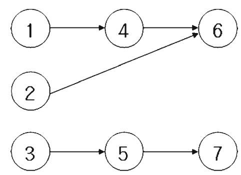

Fig. 9 A directed graph with precedence constraints

Table 3 Traveling times between each node

	Node number						
	1	2	3	4	5	6	7
Node number							
1	–	8	4	6	2	5	4
2	10	–	9	5	2	6	13
3	5	13	–	11	4	9	10
4	5	2	7	–	5	8	4
5	8	5	4	6	–	3	6
6	13	5	4	8	4	–	10
7	2	11	5	10	8	9	–

Table 4 Computation results for T-1

	Optimal value	Optimal sequence
CPLEX optimization method	26	3-5-7-1-4-2-6
He and Kusiak's method	31	1-3-5-2-4-7-6
Proposed GA approach	26	3-5-7-1-4-2-6

GA approach and two conventional approaches (CPLEX Optimization Method (2008) and He and Kusiak's method) are implemented. The computation results are shown in Table 4.

In Table 4 the proposed GA approach and the CPLEX Optimization Method locate the optimal value with the minimal traveling time of 26 (optimal sequence = 3-5-7-1-4-2-6). Unfortunately, however, He and Kusiak's method does not locate the optimal value, which means that the proposed GA approach can be used as an optimization tool for PCSPs.

Test problem 2 (T-2)

For T-2, three cases of PCSPs are considered and they are consisted of 25, 45 and 70 nodes, respectively. The detailed information (directed graphs with precedence constraints and the corresponding traveling times) of the three cases are appeared in Appendixes 1 to 4.

Table 5 shows the performance results of the TS-based representation procedure and the priority-based representation procedure when both are used in the proposed GA approach.

In terms of the optimal value in Table 5, both the priority-based representation procedure and the TS-based representation procedure have the same result for the 25 nodes. However, in the CPU time, the TS-based representation procedure is slightly quicker than the priority-based representation procedure, which means that the search scheme by the former is more efficient in locating the optimal value than that by the latter.

For the 45 nodes, in terms of the optimal value, the performance of the TS-based representation procedure is superior to that of the priority-based representation procedure, which implies that the former locates a shorter feasible sequence in terms of the optimal sequence than the latter, since the priorities assigned in each node of the latter can interrupt the finding of more feasible sequences. A similar result is also shown in terms of the CPU time. The search time by the TS-based representation procedure is quicker than that by the priority-based representation procedure, which means that the search scheme used in the former is simpler and more effective than that in the latter.

For the 70 nodes, in terms of the CPU time, the search time by the TS-based representation procedure is significantly superior to that by the priority-based representation procedure, and in terms of the optimal value the former outperforms the latter. This means that locating the optimal value by the priority-based representation procedure may be difficult or even impossible in the PCSPs with large numbers of nodes rather than the TS-based representation procedure does.

For more various comparisons with the priority-based representation procedure and TS-based representation procedure, the convergence processes of each procedure are analyzed. Figures 10 and 11 show the convergence processes for the 45 and 70 nodes until the generation number reaches to 1,000.

In Fig. 10, both the priority-based representation procedure and the TS-based representation procedure show various and fast convergence process during initial generations, However, when the generation number reaches about 650, the TS-based representation procedure has a lower fitness value than the priority-based representation procedure.

Similar result is also shown in Fig. 11. The performance of the TS-based representation procedure is more efficient than that of the priority-based representation procedure through all the generations.

The convergence results shown in Figs. 10 and 11 mean that the TS-based representation procedure has significantly better performance in locating the optimal value within the whole search space than the priority-based representation procedure.

Table 5 Computation results for T-2

		Number of nodes		
		25	45	70
Priority-based representation procedure	CPU time (second)	5.20	10.21	42.57
	Optimal value	134	223	372
	Optimal sequence	2-1-3-4-5-7-8-6-9-12-11-13-14-10-17-18-16-15-20-21-19-22-24-23-25	1-4-2-3-7-8-6-9-5-11-12-10-16-15-17-18-14-13-21-22-23-24-20-25-26-19-29-28-30-27-31-35-36-34-37-33-32-39-41-38-40-42-44-43-45	3-5-1-4-2-7-8-6-9-10-11-18-17-12-16-13-14-15-19-20-24-21-22-23-29-28-25-27-26-36-30-35-31-32-34-33-37-38-41-40-39-45-44-43-46-42-47-48-52-51-53-50-54-55-56-49-57-58-64-63-62-59-60-61-66-67-65-68-69-70
TS-based representation procedure	CPU time (second)	4.52	6.11	30.70
	Optimal value	134	219	329
	Optimal sequence	2-1-3-4-5-7-8-6-9-12-11-13-14-10-17-18-16-15-20-21-19-22-24-23-25	1-4-2-3-7-8-6-9-5-11-10-12-15-14-16-13-17-18-21-22-23-24-20-25-26-19-29-28-30-27-31-35-36-34-37-33-32-39-41-38-40-42-44-43-45	5-1-3-4-2-7-8-6-9-10-11-18-17-12-16-13-14-15-19-20-24-21-22-23-25-26-29-28-27-36-30-35-31-32-34-33-37-38-41-40-39-45-44-43-46-42-47-48-52-57-50-49-53-54-55-56-49-58-64-63-59-62-60-61-66-67-65-68-69-70

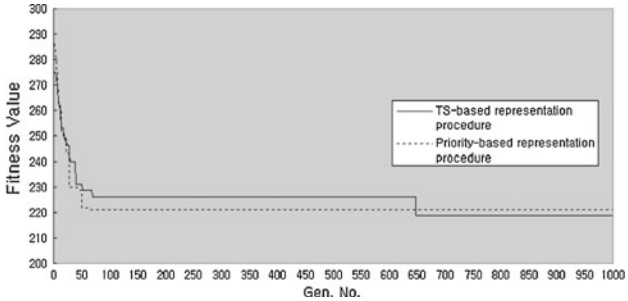

Fig. 10 Convergence process in 45 nodes

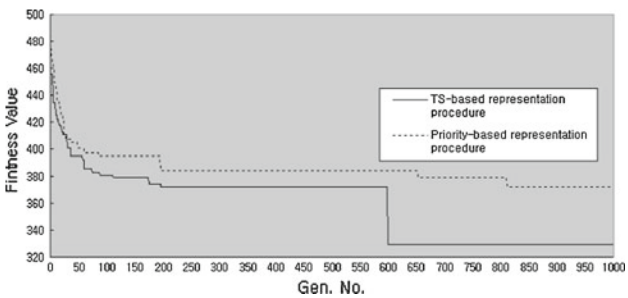

Fig. 11 Convergence processes in 70 nodes

Conclusion

This paper has proposed an efficient GA approach based on the TS-based representation procedure for solving various types of the PCSPs. By the TS-based representation procedure used in the proposed GA approach, a set of feasible sequence can be easily produced within a reasonable amount of computation time. Therefore, the proposed GA approach has produced an optimal sequence with minimal total traveling time among all feasible sequences.

For experimental comparison, two conventional algorithms (CPLEX Optimization Method and He and Kusiak's method) have been also presented and their performances have been compared with the performance of the proposed GA approach. Especially, the TS-based representation procedure has shown better performances in terms of the search speed and the ability to find optimal value than the priority-based representation procedure.

Finally we can conclude that the proposed GA approach with TS-based representation procedure has been successfully applied to various types of the PCSPs. For our future study, more large-sized PCSPs will be used for comparing the performance of the proposed GA approach with conventional GA-related approaches.

Acknowledgement This study was supported by research funds from Chosun University, 2009.

Appendix 1

See Fig. 12.

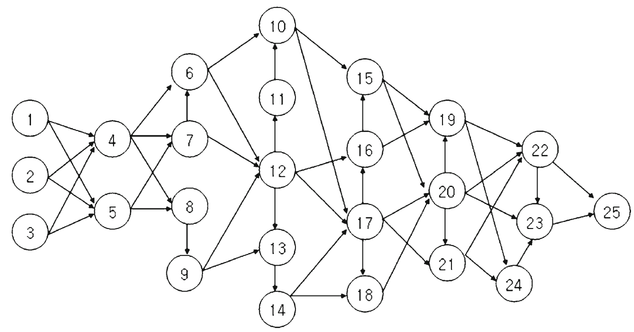

Fig. 12 PCSP with 25 nodes

Appendix 2

See Fig. 13.

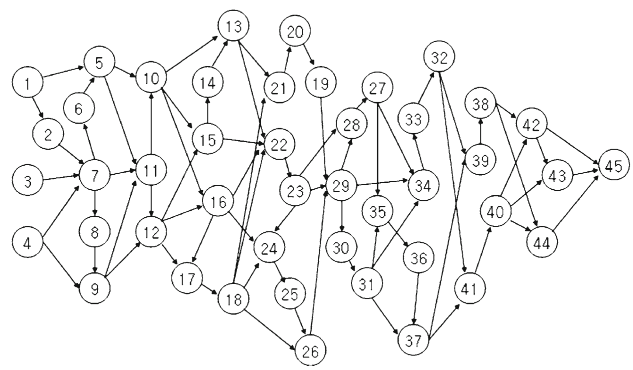

Fig. 13 PCSP with 45 nodes

Appendix 3

See Fig. 14.

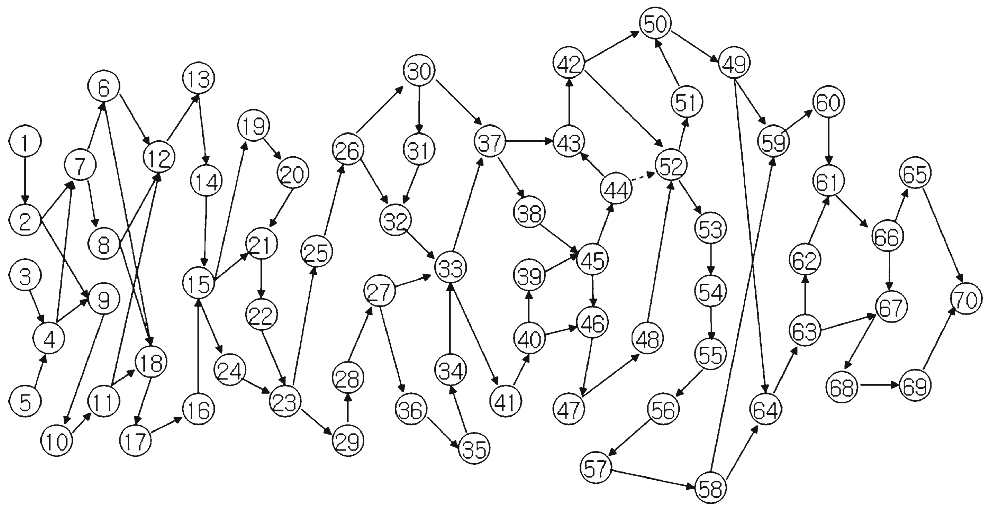

Fig. 14 PCSP with 70 nodes

Appendix 4

See Fig. 15.

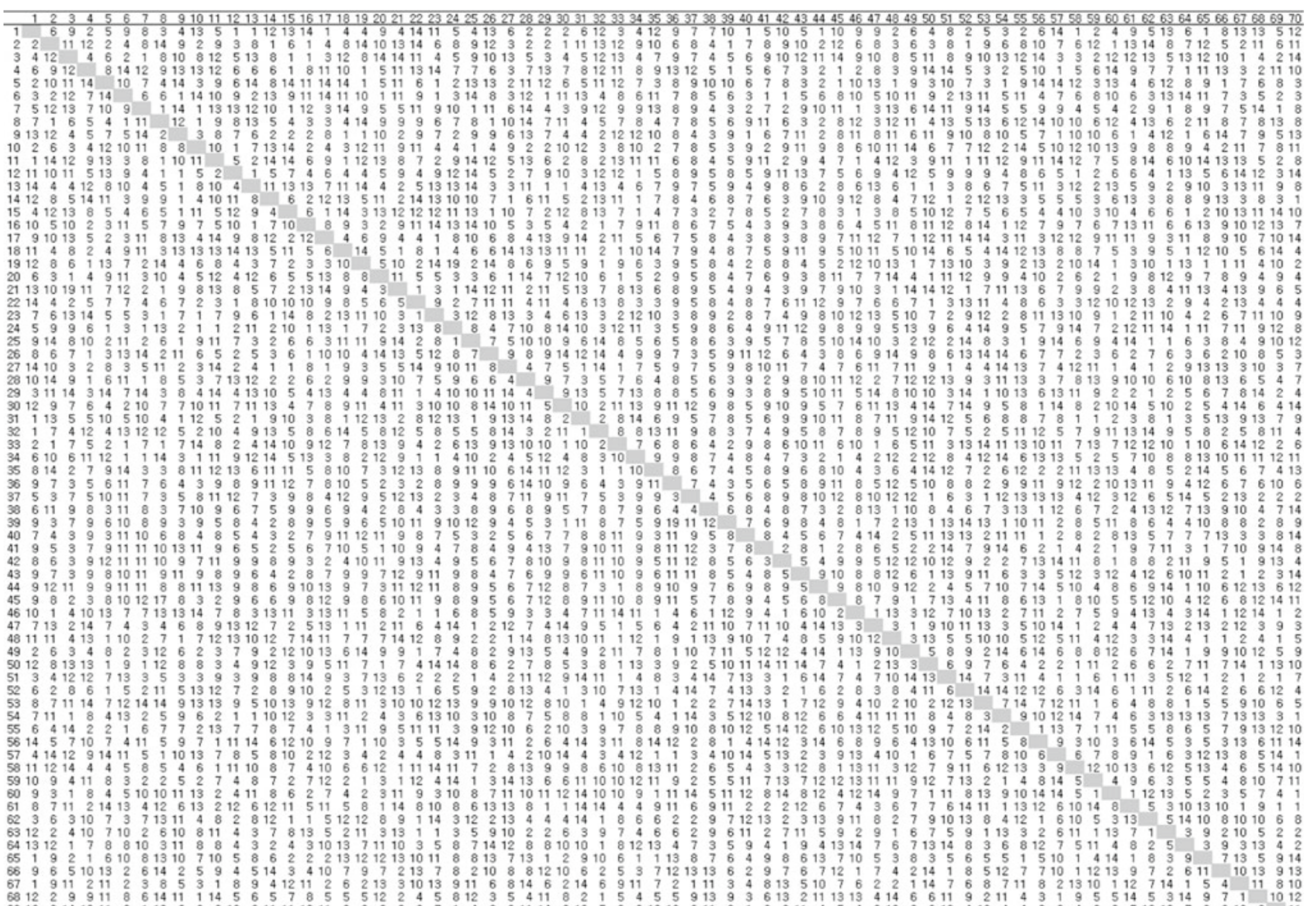

Fig. 15 Travel times in nodes 25, 45 and 70

References

Baker, K. (1974). *Introduction to sequencing and scheduling*. London: Wiley.

Chen, C. (1990). AND/OR precedence constraint traveling salesman problem and its application to assembly schedule generation. *Proceedings of 1990 IEEE International Conference on Systems, Man and Cybernetics* (pp. 560–562).

CPLEX Optimization Method. (2008). http://www.ilog.com/products/cplex/product/algorithms.cfm.

Duman, E., & Or, I. (2004). Precedence constrained TSP arising printed circuit board assembly. *International Journal of Production Research, 42*(1), 67–78.

Gen, M., & Cheng, R. (1997). *Genetic algorithms and engineering design*. London: Wiley.

Gen, M., & Cheng, R. (2000). *Genetic algorithms and engineering optimization*. London: Wiley.

Gen, M., Altiparmak, F., & Lin, L. (2006). A genetic algorithm for two-stage transportation problem using priority-based encoding. *OR Spectrum, 28*, 337–354.

He, W., & Kusiak, A (1992). Scheduling manufacturing systems. *Computers in Industry, 20*, 163–175.

Lambert, A. J. D. (2006). Exact methods in optimum disassembly sequence search for problems subject to sequence dependent costs. *Omega, 34*, 538–549.

Moon, C., Kim, J., Choi, G., & Seo, Y. (2002). An efficient genetic algorithm for the traveling salesman problem with precedence constraints. *European Journal of Operational Research, 140*(3), 606–617.

Moon, C., & Seo, Y. (2005). Advanced planning for minimizing makespan with load balancing in multi-plants chain. *International Journal of Production Research, 43*(20), 4381–4396.

Pinedo, M., & Chao, X. (1999). *Operations scheduling*. New York: McGraw-Hill.

Renaud, J., Boctor, F. F., & Ouenniche, J. (2000). A heuristic for the pickup and delivery traveling salesman problem. *Computers & Operations Research, 27*, 905–916.

Savelsbergh, M., & Sol, M. (1995). The general pickup and delivery problem. *Transportation Science, 29*, 17–29.

Su, Q. (2007). Applying case-based reasoning in assembly sequence planning. *International Journal of Production Research, 45*(1), 29–47.